CHISHOLM CATHOLIC COLLEGE

Year 11 Mathematics Methods

Test 1 2024

Total Mark
$/_{41}$

Section 1: Calculator - free

Name: Solutions	Time: 25 minutes maximum
Teacher:	Marks available: 21

Instructions:

- Answer all questions in the spaces provided.
- Calculators and notes are not permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.

Question 1 [1, 1, 2 = 4 marks]

a) Convert the following to radians. Express your answers in exact simplified form.

i) $120°$

$$\cancel{120}^{2}\cancel{0} \times \frac{\pi}{\cancel{180}_{3}\cancel{0}}$$

$$= \frac{2\pi}{3}$$ ✓

ii) $40°$

$$\cancel{40}^{2}\cancel{0} \times \frac{\pi}{\cancel{180}_{9}\cancel{0}}$$

$$= \frac{2\pi}{9}$$ ✓

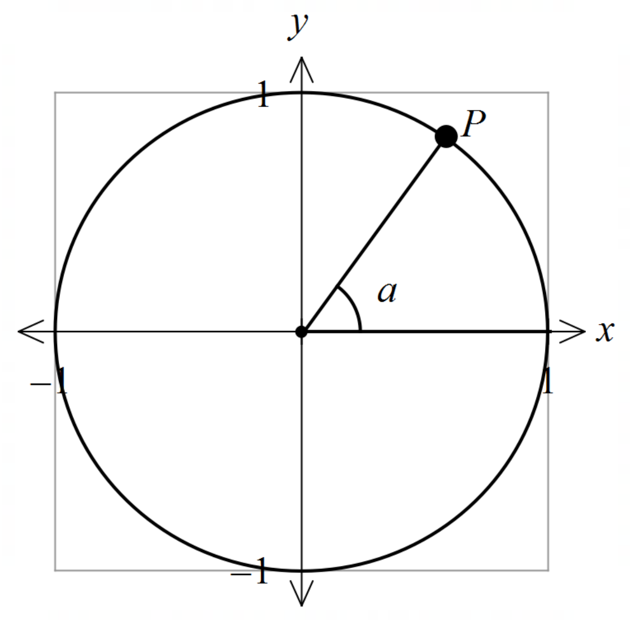
b) Convert $\frac{5\pi}{12}$ to degrees.

$$\frac{5\cancel{\pi}}{\cancel{12}_{2}} \times \frac{\cancel{180}^{3}}{\cancel{\pi}}$$ ✓

$$= 75°$$ ✓

Question 2 [1, 1, 1 = 3 marks]

The graph below shows an angle *a* in unit circle which intersects at $P(\frac{3}{5}, \frac{4}{5})$.



Determine the value of each of the following:

a) $\cos(a)$ $= \frac{3}{5}$ ✓

b) $\sin(-a)$ $= \frac{-4}{5}$ ✓

c) $\cos(180 - a)$ $= \frac{-3}{5}$ ✓

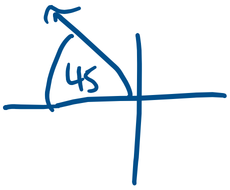
Question 3 [2, 2, 2, 2 = 8 marks]

a) Determine the exact value of the following.

i) $\sin(135°)$

$= \sin 45°$

$= \frac{\sqrt{2}}{2}$



ii) $\tan(5\pi)$

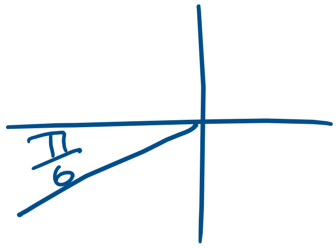
$= \tan \pi$

$= 0$

iii) $\cos(-\frac{5\pi}{6})$

$= -\cos\frac{\pi}{6}$

$= -\frac{\sqrt{3}}{2}$



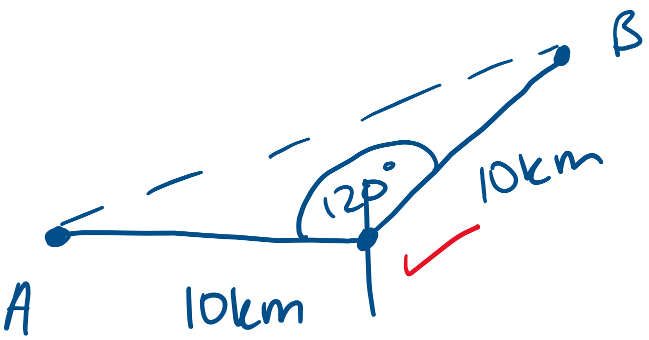
b) Determine the angle of inclination for the line given by the equation $y = \frac{\sqrt{3}}{3}x + 4$.

$\tan\theta = \frac{\sqrt{3}}{3}$

$\theta = 30°$

Question 4 [4 marks]

A bird travels 10km east of point A, then another 10km on a bearing of $030°$ to point B. Determine the exact distance between point A and point B.



$AB^2 = 10^2 + 10^2 - 2(10)(10)\cos 120°$

$= 200 - 200(-0.5)$

$= 300$

$AB = \sqrt{300}$

$= \underline{10\sqrt{3}\ \text{km}}$

Question 5 [2 marks]

Determine the obtuse angle in radians that will satisfy the equation $sinx = 0.5$.

ref angle $= \frac{\pi}{6}$.

$\therefore \quad x = \frac{5\pi}{6}$.

End of Section 1

CHISHOLM CATHOLIC COLLEGE

Year 11 Mathematics Methods

Test 1 2024

Section 2: Calculator - assumed

Name:	Time: 25 minutes minimum
Teacher:	Marks available: 20

Instructions:

- Answer all questions in the spaces provided.
- Up to 3 calculators are permitted.
- 2 double-sided A4 pages of notes are permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.
- Diagrams are not drawn to scale.

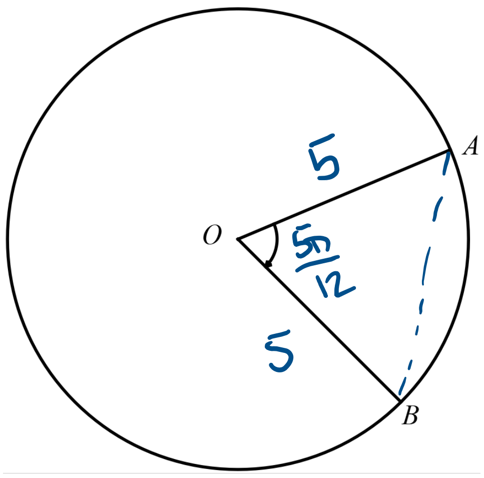
Question 6 [2 marks]

Consider the circle below where the radius is 5 m and $\angle AOB$ is $\frac{5\pi}{12}$.

Determine the length of the chord AB.

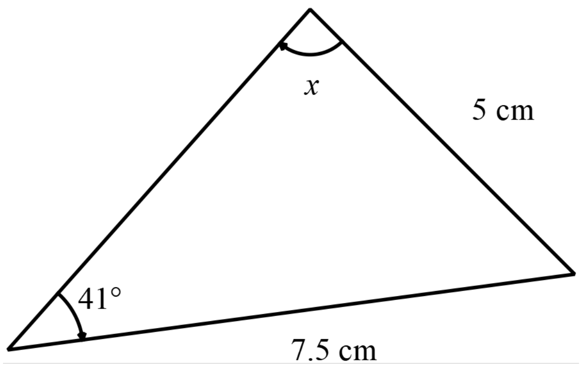
$$AB^2 = 5^2 + 5^2 - 2(5)(5)\cos\left(\frac{5\pi}{12}\right) ✓$$

$$AB = 6.09\text{ m} ✓$$



Question 7 [3 marks]

In the triangle below, determine the possible values of x, to the nearest degree.



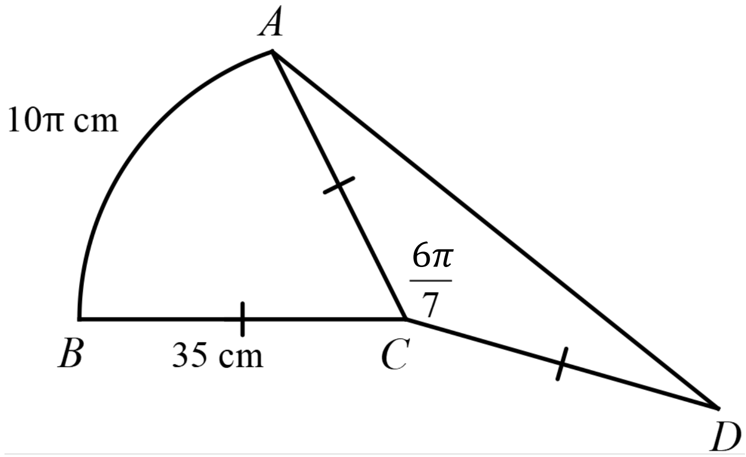
$$\frac{\sin x}{7.5} = \frac{\sin 41^\circ}{5} ✓$$

$$x = 80^\circ, \ 100^\circ$$

✓ ✓

Question 8 [1, 3 = 4 marks]

Consider the following composite shape.



a) Determine the exact angle $\angle ACB$ in radians.

$$\angle ACB = \frac{10\pi}{35}$$

$$= \frac{2\pi}{7}$$

b) Calculate the area of figure $ABCD$.

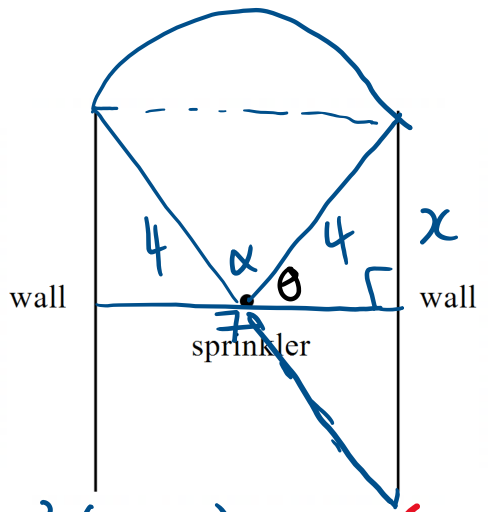
$$A = A(\text{sector}) + A(\text{triangle})$$

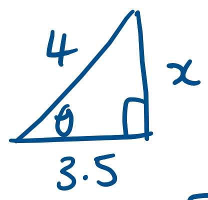
$$= \frac{1}{2}(35)^2\left(\frac{2\pi}{7}\right) + \frac{1}{2}(35)^2 \sin\left(\frac{6\pi}{7}\right)$$

$$= 815.53 \text{ cm}^2$$

Question 9 [5 marks]

A sprinkler can propel water anywhere within a 4 m radius of its location. Given that the sprinkler is located on grass, halfway between two parallel walls 7 m apart (as shown in the diagram below), determine the area of grass watered by the sprinkler.





$x = \sqrt{4^2 - 3.5^2}$

$= \frac{\sqrt{15}}{2}$.

$\theta = \cos^{-1}\left(\frac{3.5}{4}\right)$

$= 0.51$ rad.

$\alpha = \pi - 2(0.51)$

$= 2.13$ rad.

$A(\text{sector}) = \frac{1}{2}(4)^2(2.13)$

$= 17.05 \text{ m}^2$

$A(\text{triangle}) = \frac{1}{2}(3.5)\left(2 \times \frac{\sqrt{15}}{2}\right)$

$= 6.78 \text{ m}^2$

Total Area $= 2 \times A(\text{sector}) + 2 \times A(\text{triangle})$

$= \underline{\underline{47.65 \text{ m}^2}}$

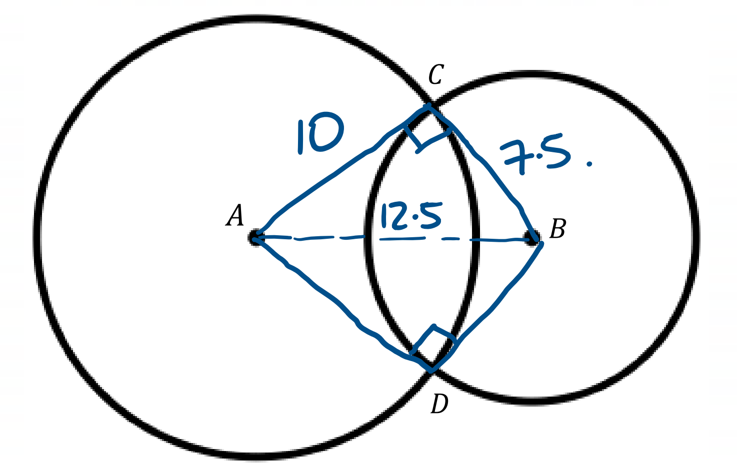
Alt Solutions Q9

$$\theta = \cos^{-1}\left(\frac{3.5}{4}\right)$$
$$= 0.51 \text{ rad}$$
$$2\theta = 1.01 \text{ rad}$$
$$A = \pi \times 4^2 - 2\left[\frac{1}{2}(4)^2(1.01 - \sin 1.01)\right]$$
$$= 47.65 \text{ m}^2$$

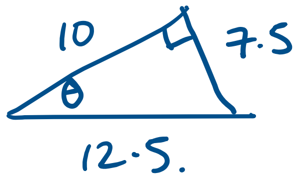
$$\theta = \cos^{-1}\left(\frac{3.5}{4}\right)$$
$$= 0.51 \text{ rad}$$
$$2\theta = 1.01 \text{ rad}$$
$$\alpha = \pi - 2\theta$$
$$= 2.13$$
$$A = 2\left(\frac{1}{2}(4)^2(2.13)\right) + 2\left(\frac{1}{2}(4)^2\sin(1.01)\right)$$
$$= 47.65 \text{ m}^2$$

Question 10 [2, 4 = 6 marks]

A circle with a centre point A has a radius of 10 cm and a smaller circle with a centre point B has a radius of 7.5 cm. The circles intersect at points C and D such that the distance AB is 12.5 cm.



a) Given $\angle ACB = \angle BDA = \frac{\pi}{2}$ radians, show that $\angle CAD = 1.29$, to two decimal places.



$\theta = \tan^{-1}\left(\frac{7.5}{10}\right)$

$= 0.6435$ rad

$\angle CAD = 2\theta$

$= 1.29$ rad.

b) Determine the overlapping area to 2 decimal places.

$\angle CBD = 2\pi - \pi - 1.29$

$= 1.85$ rad

A(overlap) = A(large seg) + A(small seg)

$= \frac{1}{2}(10)^2(1.29 - \sin 1.29) + \frac{1}{2}(7.5)^2(1.85 - \sin 1.85)$

$= 16.35 + 25.16$

$= 41.51 \text{ cm}^2$

End of Section 2